

9.3 Assessment details

This unit is assessed by means of a written examination paper. The assessment for this unit covers planning an experiment, analysing data and drawing conclusions. A laboratory is not required for this assessment.

Planning

Students will be expected to plan an experiment that is set by Pearson, although they will not be expected to carry it out.

Students may be required to:

- identify the apparatus required
- discuss calibration of instruments, e.g. whether a meter reads zero before measurements are made
- describe how to measure relevant variables using the most appropriate instrument and correct measuring techniques
- identify and state how to control all other relevant variables to make it a fair test
- discuss whether repeat readings are appropriate
- identify health and safety issues and discuss how these may be dealt with
- discuss how the data collected will be used
- identify possible sources of uncertainty and/or systematic error and explain how these may be reduced or eliminated
- comment on the implications of physics (e.g. benefits/risks) and on its context (e.g. social/environmental/historical).

Implementation and measurements

Students will be given details of an experiment carried out by an inexperienced student. Results may be included.

Students may be asked to:

- comment on the number of readings taken
- comment on the range of measurements taken
- comment on significant figures
- check a reading that is inconsistent with other readings, e.g. a point that is not on the line of a graph — students may be shown a diagram of a micrometer that is being used to measure the diameter of a wire and be expected to write down the reading to the correct number of significant figures
- comment on how the experiment may be improved, possibly by using additional apparatus (e.g. to reduce errors) — examples may include using a set square to determine whether a ruler is vertical and to aid the measurement of the extension of a spring.

Processing results

Students will be provided with a set of experimental results that were obtained by a more experienced student conducting an experiment.

Students may be expected to:

- perform calculations, using the correct number of significant figures
- plot results on a graph using an appropriate scale
- use the correct units throughout
- comment on the trend/pattern obtained
- determine the relationship between two variables or determine a constant with the aid of a graph, e.g. by determining the gradient using a large triangle
- suggest realistic modifications to reduce errors
- suggest realistic modifications to improve the experiment
- discuss uncertainties, qualitatively and/or quantitatively (students will be expected to determine the percentage uncertainty of a single measurement).

Drawing conclusions

After processing results, students may be asked to provide a final conclusion for the experiment based on their quantitative evidence.